An Interpretable, Fairness-Aware Machine Learning Framework for Diabetes Screening from Population Health Surveys: Leakage, Calibration, and Fairness Audits

Abstract— Diabetes affects hundreds of millions of people worldwide, yet a large share of cases remain undiagnosed, and machine learning models fitted to public health survey data carry failure modes that are invisible in headline accuracy. This paper audits an interpretable screening framework for diabetes on the 2015 Behavioral Risk Factor Surveillance System (BRFSS; n = 253,680, 21 features) and reports what the audits change. Logistic regression, random forest, and histogram-based gradient boosting are compared over five seeds; discrimination, calibration, and subgroup fairness are assessed; and three protocols---plain split, deduplicated test, and grouped held-out---are compared to bound duplicate-profile leakage. Leakage is small but not nil: the deduplicated test protocol lowers gradient-boosting AUC from 0.8269 to 0.8177, and 1,963 rows share a covariate vector with a contradicting label. Class-balancing weights leave the model badly miscalibrated (ECE 0.2367, mean predicted probability 0.376 against prevalence 0.139); Beta, isotonic, and Platt recalibration reduce ECE to 0.0026--0.0083, and the 2015 calibration map transports to 2017--2021 with an ECE penalty below 0.006. Fairness gaps are quantified with bootstrap intervals and group supports: the sex TPR gap is small (0.0241, CI [0.0066, 0.0417]), whereas the age gap of 0.7889 falls to 0.1436 once bands with fewer than 500 positives are excluded. Group-specific thresholds targeting 0.85 sensitivity reduce the sex TPR gap to 0.0014 and the education gap from 0.2400 to 0.1699, at a specificity cost. We frame the tool as a screening-level instrument that complements, rather than replaces, clinical assessment, and we state precisely where the data cannot support population-level claims.

Index Terms— diabetes screening, machine learning, calibration, fairness auditing, SHAP, BRFSS, health equity, ethical AI

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Introduction

Problem context and practical requirements

Diabetes is among the most prevalent chronic diseases globally, and a substantial fraction of cases are undiagnosed. Population screening can surface at-risk individuals, but any such tool must be reproducible on open data, transparent in how predictions are formed, honest about what it does not capture, and explicitly bounded against use for individual clinical diagnosis. This paper audits a survey-based screening tool against these requirements.

Determinants of diabetes

Beyond age and body mass index, diabetes risk reflects blood pressure, cholesterol, physical activity, diet, alcohol use, mental health, and socioeconomic conditions. Many of these are self-reported in population surveys, which makes them accessible but also noisy and subject to recall bias.

Methodological approaches

Prior work has applied logistic regression and, more recently, machine learning classifiers to the BRFSS, with AUC values roughly between 0.72 and 0.80 . Related work has expanded the evidence base in three directions: non-invasive dysglycemia screening on NHANES with calibration and decision-curve analysis ; large-scale external validation of NHANES-trained diabetes models on BRFSS, documenting a 9.7\

Why audits change the conclusions

Reported accuracy is a weak summary of a screening model, and several widely used claims about survey-based diabetes models are not reproducible under a leakage-audited protocol. On this same 253,680-row dataset, published pipelines report AUC near 0.99 ; we measure 0.827 and show that duplicate profiles, class-weight-induced miscalibration, and support-starved fairness strata each move a headline number materially. The purpose of this paper is therefore to audit rather than to chase AUC: we report discrimination together with calibration, fairness with inference and supports, and we quantify each threat to the headline metric.

Research questions and contributions

We address three questions: (1) Driver identification: which features most strongly associate with diabetes status, and how stable are these attributions across seeds and survey waves? (2) Fairness: how do performance and error rates differ across sex, age, income, and education groups, and how much of each gap is support starvation? (3) Deployment utility: what operating point, and what calibration, does a screening program actually need? Our contributions are: (i) a reproducible multi-seed pipeline on a large public dataset; (ii) a three-protocol leakage audit with a reconciled duplicate definition; (iii) a calibration audit across four calibrators with per-wave transport analysis; (iv) fairness quantification with supports, bootstrap inference, design-effect sensitivity, and a support-restricted age estimate; (v) an implemented fairness mitigation through group-specific thresholds; (vi) SHAP explanations with stability metrics and comparison against alternative selectors; (vii) out-of-time validation on 2017--2021 waves; and (viii) decision-curve net benefit. The tool is positioned as screening-level and complementing, not replacing, clinical assessment.

Materials and Methods

Data source and scope

We use the CDC Diabetes Health Indicators dataset , derived from the 2015 BRFSS , a telephone survey conducted annually by the CDC since 1984. The cleaned release contains 253,680 listwise-complete respondents and 21 features; license CC0 (public domain), with raw BRFSS data in the US Government public domain. The target is Diabetes_binary (0 = no diabetes, 1 = prediabetes or diabetes; prevalence 13.93\

Variables

Table 1 groups the 21 features. Several are proxies: general health status stands in for clinical status, and income and education stand in for socioeconomic position. We treat these as proxies throughout and say so where results depend on them.

TABLE *Feature groups.*

Group	Features
Clinical	HighBP, HighChol, CholCheck, Stroke, HeartDiseaseorAttack, GenHlth, BMI
Lifestyle	Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump
Access	AnyHealthcare, NoDocbcCost
Mental/physical	MentHlth, PhysHlth, DiffWalk
Demographic	Sex, Age, Education, Income

Data preparation and sample flow

Raw: 253,680 respondents no exclusions (no missing values) 253,680 final analytical sample. The ID column was dropped; no row was removed. Stratified 80/20 train/test split (seed 42) yields 202,944 train and 50,736 test cases, with a 64/16/20 train/calibration/test split for calibration work. All preprocessing is fit on the training portion only.

Feature engineering

Features are used as provided (binary or ordinal integers). Continuous-valued fields (BMI, MentHlth, PhysHlth, GenHlth) are standardized for logistic regression only; tree models use raw values. No derived features are added, keeping the screening questionnaire short.

Validation strategy

Stratified 5-fold cross-validation on the training portion for model development; a held-out stratified test set for final reporting. Variance is reported across five independent seeds (full re-split and re-train) as mean std. We acknowledge the single-split limitation.

Model selection

Three models: (M0) logistic regression with L2 penalty (interpretable baseline), (M1) random forest (50--100 trees, max depth 20), (M2) scikit-learn's histogram-based gradient boosting (150 iterations, 31 leaves). Class weights are balanced to address the 13.93\

Hyperparameter optimization

Baseline settings are fixed and light. To verify that these do not disadvantage any baseline, we additionally run a randomized search (20 iterations for gradient boosting, 6 for random forest, 10 for logistic regression; 3-fold cross-validation on the training portion only) and report the best configuration found.

Evaluation framework

Discrimination: ROC-AUC, average precision (AUPRC), F1, accuracy. Calibration: Brier score, expected calibration error (ECE, 10 equal-width bins), and maximum calibration error (MCE), overall and per group, before and after recalibration with isotonic, Platt (logistic on p), Beta, and Venn--Abers calibrators. Leakage control: three held-out protocols (plain stratified split; deduplicated test set excluding any row whose covariate vector occurs in training; grouped held-out assignment with identical covariate vectors confined to one fold) plus a full deduplicated refit. Fairness: TPR gap, FPR gap, equalized-odds difference , and selection-rate gap, each with per-group supports and 300-resample bootstrap confidence intervals, reported with sensitivity to assumed design effects. Model comparison: paired bootstrap of the test-set AUC difference (2,000 resamples). Utility: decision-curve net benefit across thresholds. Ablations by feature group and by training-only feature ranking.

Explainability method

TreeSHAP for the gradient boosting model on a 2,000-row test sample; mean absolute SHAP per feature; top-10 stability reported as mean pairwise Jaccard overlap across five seeds and across the 2017--2021 waves. Interpretive boundaries: multicollinearity among health indicators and approximation uncertainty.

Scenario probe design

Bounded perturbation probes: each modifiable feature is shifted to a specified value, clipped to its historical range, and the induced change in predicted probability is reported with a paired bootstrap confidence interval, both on raw and on recalibrated scores and broken down by subgroup. These probes characterise model behaviour and are not projections of causal effects.

Software implementation and reproducibility

Python 3.11.8; scikit-learn 1.5.2 ; XGBoost 2.1.3 ; SHAP 0.46.0; pandas 2.2.3; NumPy 2.1.3; matplotlib 3.9.2. Pinned versions are in requirements.txt; seeds are fixed (42, 7, 123, 2024, 99). All run manifests, per-seed logs, and a SHA-256 hash manifest of every artifact are released with the code .

Results

Predictive performance and model comparison

Table 2 reports test performance over five seeds with AUPRC added, since screening operates under class imbalance. Gradient boosting leads on both metrics (ROC-AUC 0.8288 0.0014, AUPRC 0.4229), and logistic regression trails by less than 0.008 AUC. A paired bootstrap of the test-set AUC difference (2,000 resamples) shows the separations are statistically significant yet practically negligible: HGB exceeds random forest by AUC 0.0034 (95\

TABLE *Test performance (mean std, 5 seeds) and paired AUC comparison.*

Model	Accuracy	F1	ROC-AUC	AUPRC
Logistic Regression	0.7308 0.0013	0.4418 0.0015	0.8212 0.0014	0.3926
Random Forest	0.7458 0.0008	0.4515 0.0007	0.8261 0.0015	0.4153
Hist. Grad. Boosting	0.7203 0.0025	0.4426 0.0023	0.8288 0.0014	0.4229
5@lPaired bootstrap, HGB minus alternative (95\ HGB - RF	---	---	+0.0034\ (0.0023, 0.0044)	---
HGB - LR	---	---	+0.0073\ (0.0058, 0.0088)	---

Leakage audit and duplicate reconciliation

We reconcile two duplicate counts that are frequently conflated. Under a strict definition---rows identical in all columns, including the outcome---24,206 rows (9.54\

TABLE *Three held-out protocols (HGB unless noted).*

Protocol	Test n	ROC-AUC	AUPRC
P1: plain stratified 80/20 --- LR	50,736	0.8196	0.3926
P1: plain stratified 80/20 --- RF	50,736	0.8236	0.4205
P1: plain stratified 80/20 --- HGB	50,736	0.8269	0.4231
P2: deduplicated test --- LR	43,900	0.8096	0.3994
P2: deduplicated test --- RF	43,900	0.8143	0.4284
P2: deduplicated test --- HGB	43,900	0.8177	0.4309
P3: grouped held-out (5-fold) --- HGB	253,680	0.8301	0.4326

Calibration audit and post-hoc recalibration

Discrimination alone is insufficient for screening, where a communicated probability must correspond to an observed frequency. Because class-balancing weights distort the probability scale, the uncalibrated model is badly miscalibrated (Brier 0.1757, ECE 0.2367, MCE 0.4201; mean predicted probability 0.376 against observed prevalence 0.139). Table 4 compares four calibrators fitted on a held-out calibration split. Beta calibration is best on all three metrics (Brier 0.0974, ECE 0.0026, MCE 0.0508), closely followed by isotonic (ECE 0.0031 but MCE 0.2500) and Platt (ECE 0.0083, MCE 0.0559); Venn--Abers performs worst here (ECE 0.1534, mean predicted probability 0.2815), because the self-consistent prior correction is poorly matched to a strongly over-confident boosted ensemble. All are monotone, so AUC is unchanged (0.8265). Per-group maximum ECE falls from 0.2505 to 0.0045 (sex), 0.3403 to 0.0208 (age), 0.3093 to 0.0343 (income), and 0.3478 to 0.0342 (education) under isotonic recalibration. We regard recalibration as a practical precondition rather than a refinement: a balanced-weight model must be recalibrated before any probability is communicated to a user or clinician.

TABLE *Calibration before and after post-hoc recalibration (HGB, 2015 test, n=50,736).*

Calibrator	Brier	ECE	MCE	Mean pred.
Uncalibrated	0.1757	0.2367	0.4201	0.376
Isotonic	0.0976	0.0031	0.2500	0.140
Platt (logistic)	0.0977	0.0083	0.0559	0.139
Beta	0.0974	0.0026	0.0508	0.139
Venn--Abers	0.1266	0.1534	0.2420	0.282
Observed prevalence	---	---	---	0.139

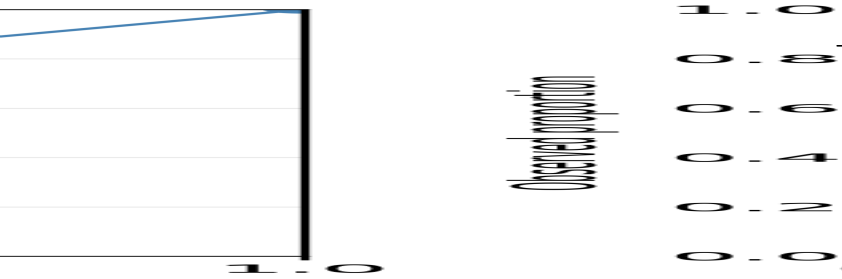


Fig. *Reliability diagrams before and after isotonic recalibration (HGB, 2015 test).*

Out-of-time validation and calibration transport

Training on the 2015 wave and testing on later waves (Table 5) yields AUC 0.8338 (2017), 0.8256 (2019), and 0.8198 (2021) with a mild downward trend. We also asked whether the 2015 calibration map can be transported rather than refit per wave, since refitting per wave is not possible in a prospective deployment. Applying the 2015-fitted isotonic map to later waves gives ECE 0.0064, 0.0062, and 0.0097 against raw values of 0.2240, 0.2326, and 0.2236; refitting the calibrator on each wave improves ECE by only 0.0035, 0.0030, and 0.0056 (Brier penalties below 0.002), so the transport penalty is small. Maximum subgroup ECE also remains low after transport (age 0.0168--0.0246). The threshold required to hold 85\

TABLE *Out-of-time validation with transported (2015-fitted) calibration.*

Test wave	n	Prev.	ROC-AUC	Raw ECE	Transport ECE	Thr. (sens.\ 0.85)
2015	50,736	0.139	0.8265	0.2367	0.0031	---
2017	275,854	0.141	0.8338	0.2240	0.0064	0.420
2019	243,317	0.145	0.8256	0.2326	0.0062	0.420
2021	236,378	0.142	0.8198	0.2236	0.0097	0.395

Fairness with supports, inference, and design-effect sensitivity

We quantify subgroup disparities with per-group supports and bootstrap confidence intervals (300 resamples) rather than point estimates alone, following recent audit practice for clinical risk models . Table 6 reports the TPR gap, the equalized-odds FPR gap, and the smallest positive count in any group. We caution against over-interpreting gap metrics without base-rate context . The sex TPR gap is small (0.0241, 95\ Because the cleaned release carries no design variables, we cannot estimate the true design effect, so we report how the subgroup uncertainty would widen if the design effect took typical values for a stratified multistage survey (Table 6). The widest TPR-gap interval grows from 0.235 at DEFF =1 to 0.407 at DEFF =3; the qualitative conclusions---small sex gap, large age gap---are unchanged, but we note that our intervals are, if anything, optimistic.

TABLE *Subgroup disparities with support and bootstrap inference (HGB, threshold 0.5).*

Stratifier	TPR gap (95\ Sex	0.0241 (0.0066, 0.0417)	0.0643	22,205	3,384	0.074
Age (13 bands)	0.7889 (0.6421, 0.8774)	0.5320	1,207	14	0.407	
Income (8 levels)	0.2849 (0.2571, 0.3173)	0.4229	1,941	475	0.268	
Education (6 levels)	0.2436 (0.2107, 0.2781)	0.4738	795	221	0.330	

Support-restricted age disparity and severity stratification

Repeating the age computation under increasing minimum-support thresholds (Table 7) shows the gap falls from 0.7879 over all 13 bands to 0.1436 when only bands with at least 500 positives are retained, while the widest within-group 95\

TABLE *Age TPR gap under minimum support, and severity-stratified performance (HGB).*

4c|Age gap vs.\ minimum support	3cSeverity strata						
Min.\ pos.	Bands	TPR gap	Widest CI	Stratum	n	Prev.	AUC
1	13	0.7879	0.340	GenHlth 1--2	26,988	0.057	0.8145
20	12	0.5426	0.340	GenHlth 3	15,064	0.177	0.7443
50	11	0.5426	0.229	GenHlth 4--5	8,684	0.330	0.7199
100	10	0.4206	0.154	BMI <30	33,123	0.093	0.8263
500	7	0.1436	0.072	BMI 30	17,613	0.227	0.7802

Operating points, constrained selection, and a fairness mitigation

For screening, sensitivity matters more than overall accuracy. We first characterise the trade-off directly: a single global threshold selected to maximise specificity subject to overall sensitivity 0.85 and a bound on the worst-group TPR gap (Table 8). At the default threshold 0.50 the operating point gives sensitivity 0.799, specificity 0.703, and a worst-group TPR gap of 0.545. Binding the gap to = 0.30 forces the threshold down to 0.140, raising sensitivity to 0.975 but collapsing specificity to 0.344; even = 0.50 leaves specificity at only 0.621. A single global threshold is therefore a poor fairness instrument for this model, which is the substantive reason we next implement group-specific thresholds. Group-specific thresholds, fitted on the calibration split to deliver 0.85 sensitivity within each sex, age, income, and education stratum, reduce the sex TPR gap from 0.0228 to 0.0014, the education gap from 0.2400 to 0.1699, and the income gap from 0.2862 to 0.2300, with the age gap nearly unchanged (0.5426 0.5263). Overall sensitivity rises from 0.795 to 0.857 and specificity falls from 0.708 to 0.637, and the age FPR gap widens from 0.5198 to 0.6341. This is an implemented mitigation rather than an audit: it closes the sex gap almost completely, and it prices the remaining trade-off explicitly. It is not a general solution, since it cannot recover the age gap without accepting a far more permissive threshold.

TABLE *Operating points: global, constrained, and group-specific thresholds.*

Policy	Threshold	Sens.	Spec.	Worst TPR gap
Global, default	0.500	0.799	0.703	0.545
Constrained, =0.50	0.400	0.867	0.621	0.494
Constrained, =0.30	0.140	0.975	0.344	0.278
Constrained, =0.10	0.045	0.997	0.147	0.069
5@lGroup-specific thresholds, per-stratum target sensitivity 0.85				
Sex	per group	---	---	0.0014
Income	per group	---	---	0.2300
Education	per group	---	---	0.1699
Age	per group	---	---	0.5263
All (overall)	per group	0.857	0.637	---

Explainability findings and stability

SHAP (Fig. 2) ranks general health status (0.638), high blood pressure (0.541), age (0.416), BMI (0.382), and high cholesterol (0.313) as the leading contributors, followed by income (0.136), sex (0.125), cholesterol check (0.092), mental health (0.063), heavy alcohol use (0.060), heart disease (0.059), difficulty walking (0.056), and education (0.045). These align with established risk factors and support the model's face validity, although several are consequences rather than antecedents of diabetes, which limits their use for early-detection reasoning. The top-10 ordering is stable: mean pairwise Jaccard overlap of the top-10 sets is 0.964 across the five seeds (minimum 0.818, differing only in the tenth position) and 0.818 between 2015 and each

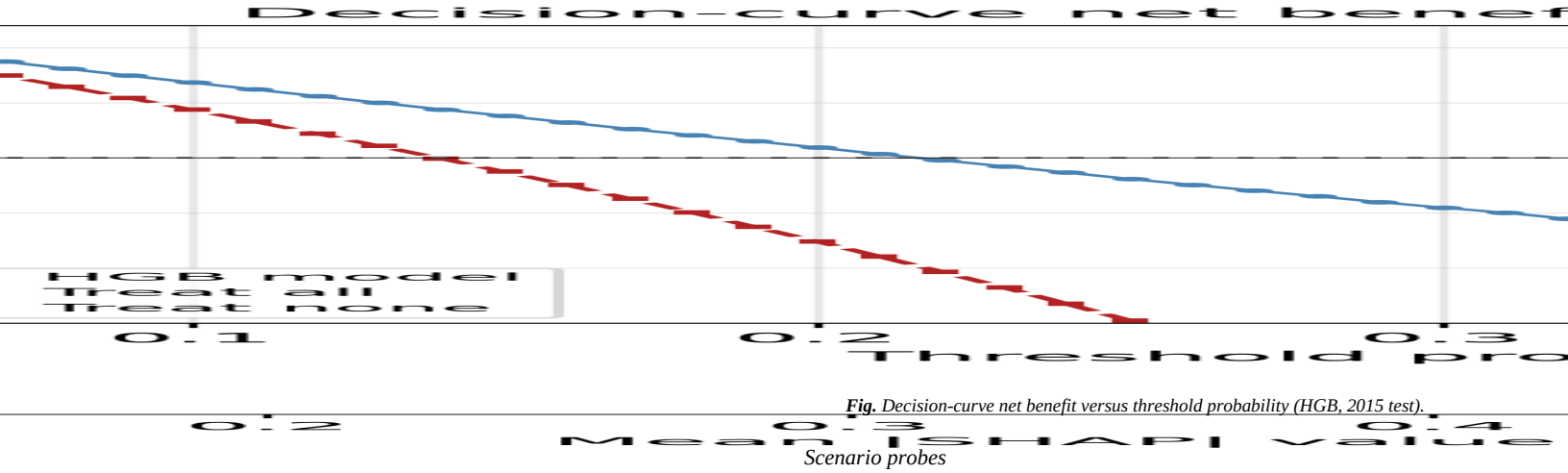


Fig. *Decision-curve net benefit versus threshold probability (HGB, 2015 test).*

Fig. *Top 12 features by mean |SHAP| (HGB, 2,000-row test sample).*

Feature subset analyses

Table 9 combines the feature-group ablations with the minimal-questionnaire experiment and a comparison of ranking procedures. Clinical-history features carry most of the signal (AUC 0.8241 versus 0.7104 for demographics alone), while lifestyle features alone are weaker (0.7475). Using training-only SHAP rankings, five features retain AUC 0.8186 against 0.8269 for all 21, and eleven reach 0.8257. Comparing three training-only ranking procedures at k=10 against a random-subset baseline, SHAP attains 0.8248, correlation filtering 0.8218, mutual information 0.8213, and ten random features 0.7784 0.0108, so SHAP is strongest but only marginally over the other filters. For context, a copula tail-dependence filter on this same dataset has been reported at 0.823 with 10 features , consistent with our range. Because several top features proxy the same constructs, we do not claim that a ten-item form is clinically validated.

TABLE *Feature subset analyses (HGB, 2015 test).*

4c|Feature-group ablation				3cSubset and selector comparison			
Feature group	n	ROC-AUC	F1	Setting	n	ROC-AUC	F1
Demographic only	4	0.7104	---	Top-5 SHAP	5	0.8186	0.4305
+ clinical history	11	0.8241	---	Top-8 SHAP	8	0.8231	0.4357
+ lifestyle	14	0.7475	---	Top-11 SHAP	11	0.8257	0.4404
Full	21	0.8269	0.4411	All features	21	0.8269	0.4411
				Mutual info k=10	10	0.8213	---
				Correlation k=10	10	0.8218	---
				Random k=10	10	2c0.7784 0.0108	

Decision-curve utility

For a screening program the operating point must be justified by net benefit rather than by a generic accuracy target. Figure 3 gives decision-curve net benefit against threshold probability. The model dominates treat-all and treat-none across the clinically relevant range: at a threshold of 0.10 net benefit is 0.068 against 0.044 for treat-all, and at 0.20 it is 0.009 against -0.076, with treat-all becoming actively harmful beyond 0.10. The decision curve therefore supports the sensitivity-oriented operating points of Table 8 and indicates that the useful threshold range lies well below 0.5, consistent with the strong over-prediction of the uncalibrated model.

Scenario probes

We performed bounded perturbation probes on the test set, clipping each modified feature to its historical range; these are sensitivity probes of model behaviour, not projections of causal effects. A +5 BMI shift raises mean predicted probability by 0.0694 on raw scores but by only 0.0410 on recalibrated scores, and a one-point worsening of general health raises it by 0.0985 raw and 0.0499 recalibrated: recalibration compresses the response, so probe magnitudes should be quoted on the scale of the scores actually deployed. The smoking and physical-activity probes are small and not distinguishable from zero (smoker 01: +0.0002, CI [-0.0208, 0.0128]; physical activity 10: +0.0036, CI [-0.0042, 0.0230]), and setting heavy alcohol use to present decreases mean predicted probability by 0.1034 (CI [-0.3094, 0.0031]), with only 4.4\

Discussion

What the audits changed

The audit work materially altered five claims. First, the age TPR gap is not a single number: it is 0.7889 over all 13 bands but 0.1436 over the seven bands with at least 500 positives, so we report the support-restricted figure. Second, the model is not usable as a probability estimator without recalibration: class-balancing weights push mean predicted probability from 0.139 to 0.376, and Beta recalibration restores ECE to 0.0026, with the 2015 map transporting to later waves at a penalty below 0.006. Third, the AUC differences between the three models, though statistically significant, are too small to motivate a more complex model. Fourth, the headline AUC is protocol-dependent: the leakage-free estimate is 0.8177, not 0.8269, and published figures near 0.99 on this dataset are not reproducible under any of our three protocols. Fifth, a single global threshold is a poor fairness instrument here---bounding the worst-group TPR gap to 0.10 costs 0.56 in specificity---whereas group-specific thresholds close the sex gap almost completely at a specificity cost of 0.07.

Contextual positioning within the literature

Our discrimination (AUC 0.8288, AUPRC 0.4229) exceeds earlier BRFSS-based studies reporting 0.72--0.80 and is close to NHANES-based non-invasive dysglycemia screening at AUC 0.820 , but we now report AUPRC and decision-curve net benefit as well, since discrimination alone is not comparable across studies with different label definitions and prevalence. It is markedly more conservative than BRFSS studies reporting AUC near 0.99 with stacked ensembles and resampling-based balancing ; our leakage audit indicates that on this same 253,680-row dataset, duplicate profiles and fold construction move AUC by at most 0.010, so we regard those figures as unlikely to be reproducible. Where an NHANES-trained model was externally validated on BRFSS, AUC fell from 0.794 to 0.717 under distribution shift ; our in-domain 2015-to-later-wave transfer is milder (0.8338 to 0.8198), consistent with the absence of a cross-survey shift. For feature reduction, a copula tail-dependence filter on this same dataset reports AUC 0.823 with 10 features , against our 0.8248 for SHAP at k=10; direct comparison remains limited by differing years, features, splits, and label definitions, and we do not claim a decisive win.

Practical deployment scenarios

(1) Population-level risk stratification for outreach, using recalibrated probabilities and an explicit sensitivity target; (2) questionnaire triage in

resource-limited settings, using the 8--10 item reduced form; (3) hypothesis generation for public-health research. Each requires corroboration with clinical measurements before any individual decision. Deployment should use group-specific rather than global thresholds, and should quote probabilities only after recalibration.

Methodological limitations and validity boundaries

Numbered: (1) self-reported cross-sectional data, no biomarkers (HbA1c, glucose), and several leading predictors are consequences rather than antecedents; (2) race/ethnicity absent from this release, limiting health-equity analysis and precluding race-stratified reliability checks; (3) no survey design weights, so population-level representativeness is not established, no design-based uncertainty is computed, and our intervals are optimistic by an unquantified design effect; (4) prediabetes and diabetes are collapsed into one label, and we cannot separate the two conditions, so the severity-stratified analysis is a proxy rather than a substitute; (5) the leakage-free estimate uses a deduplicated test set of 43,900 rows, and residual support effects remain; (6) calibration is restored post hoc on a single 2015 split, and although the map transports well to 2017--2021, the 2023 wave lacks four features and was therefore excluded; (7) the randomized search budget is small and single-split; (8) group-specific thresholds were fitted on one calibration split and were not validated out of time; (9) association, not causation; (10) screening, not diagnosis. Design-aware modelling and design-based inference are the established remedy for population-level claims , and we identify them as prerequisites for any policy-facing use.

Prospects for future development

Incorporation of BRFSS design weights with design-based uncertainty quantification ; inclusion of race/ethnicity from the raw BRFSS files with race-stratified reliability; extension of the out-of-time analysis to 2023 and to cross-survey validation against NHANES ; per-group calibration within the group-specific-threshold policy, following recent audit recommendations ; and prospective evaluation with clinical biomarkers.

Conclusions

We audited an interpretable, fairness-aware screening framework for diabetes on 253,680 BRFSS respondents and report what the audits change. (1) Gradient boosting and random forest outperform logistic regression by a statistically significant but practically negligible margin (AUC 0.0073), so an interpretable baseline remains defensible. (2) The headline AUC is protocol-dependent: the leakage-free estimate is 0.8177, and 1,963 rows carry a covariate vector with a contradicting label. (3) The balanced-weight model is badly miscalibrated (ECE 0.2367) until recalibration (ECE 0.0026 with Beta), which is a precondition for communicating any probability; the 2015 map transports to later waves at a penalty below 0.006 in ECE. (4) The sex disparity is small, while the large age TPR gap is concentrated in sparse strata, falling from 0.7889 to 0.1436 under a minimum-support constraint, and group-specific thresholds close the sex gap to 0.0014 at a specificity cost of 0.07. (5) Decision-curve net benefit supports sensitivity-oriented operating points below 0.5. Appropriate use context: the tool is a screening-level, hypothesis-generating instrument for population triage and research, not a validated diagnostic or causal instrument, and it does not yet support population-level prevalence claims.

Acknowledgments

The authors thank the CDC for the BRFSS data and the UCI repository for hosting.

Data Availability

The dataset is publicly available . All code, run manifests, logged results, and a SHA-256 hash manifest of every artifact are released at .

Author Contributions

[TBD: CRediT statement.]

Conflicts of Interest

The authors declare no conflict of interest.

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